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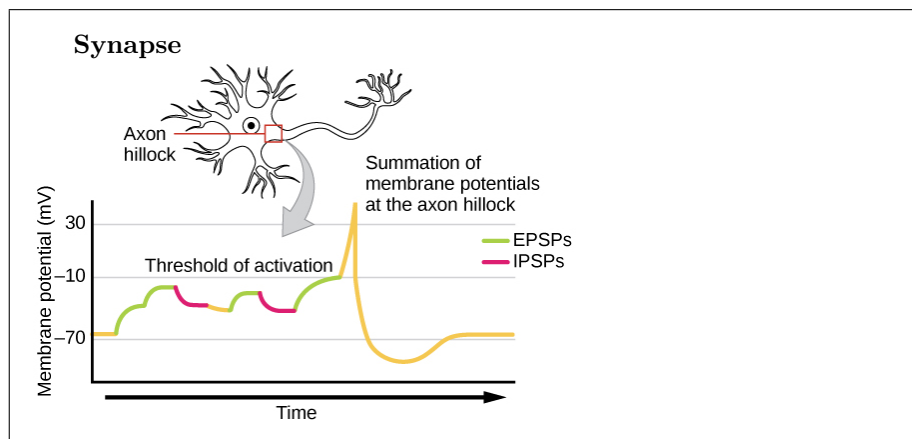
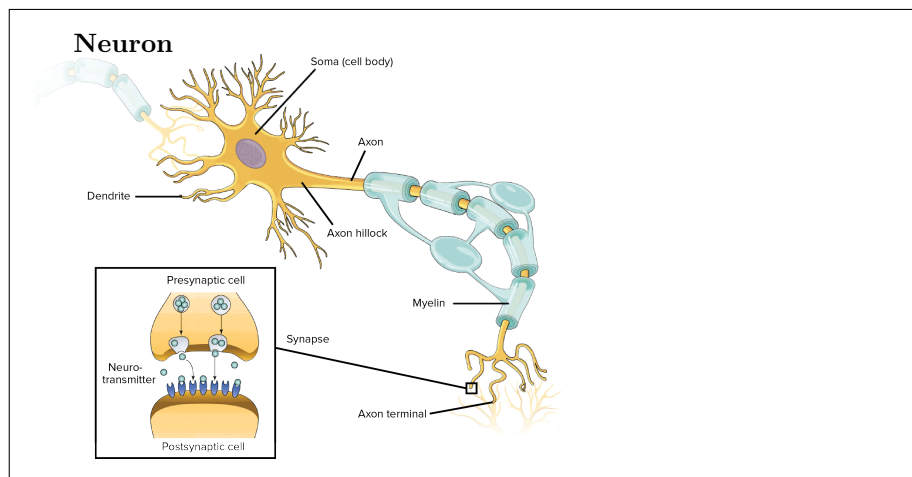
What to expect in the course

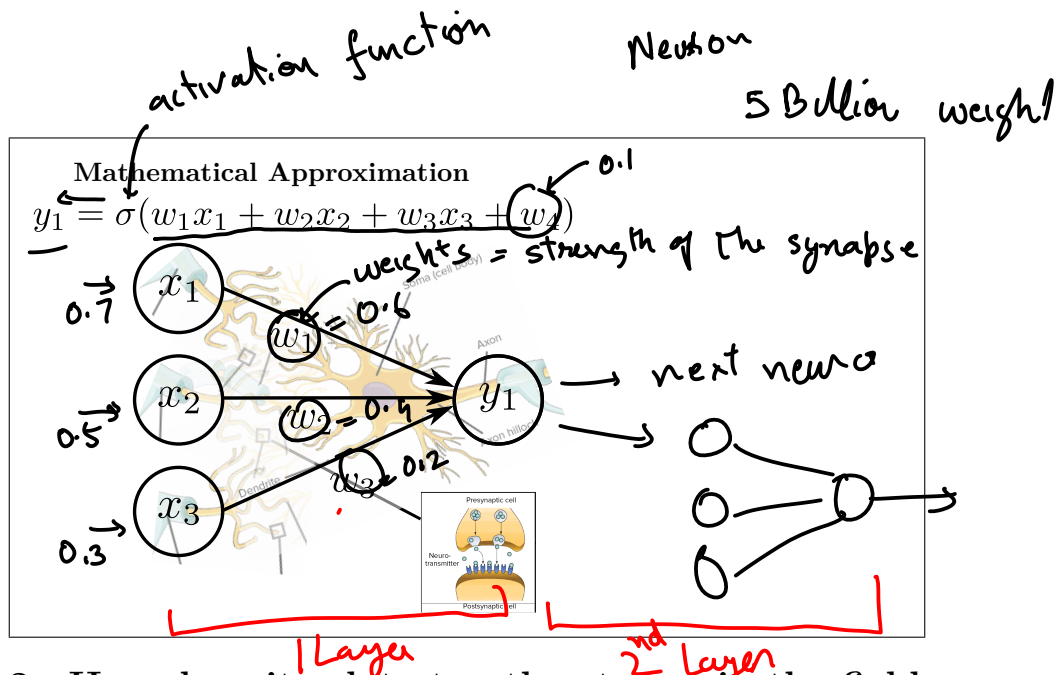
Vikas Dhiman

Tuesday 2nd September, 2025



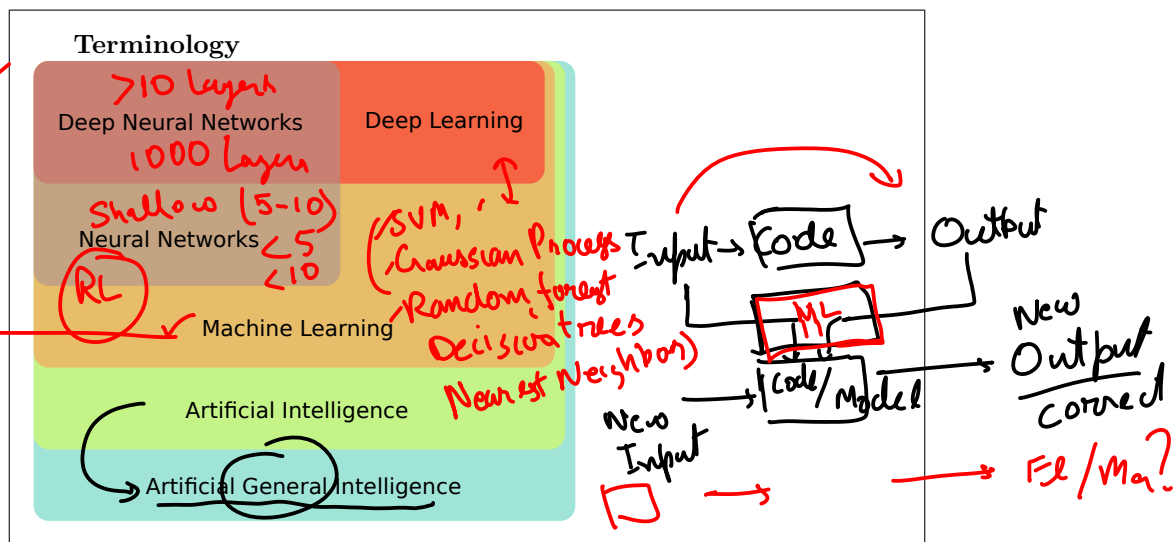
1 What is a Neural Network?





1 Layer 2nd Layer

2 How does it relate to other terms in the field



2.1 Comparison with other courses

Other courses

- ECE 491/591: Deep Learning (Dr Yifeng Zhu)
 - We will dive deeper on backpropagation. We will implement your on Pytorch library. *inside Pytorch*

1-layer NN / Linear models
Perceptron

- We are more theoretical, less towards application than ECE 491.
- COS 475/575: Machine Learning (Chaofan Chen)
 - There are other kinds of ML algos (SVM, Gaussian Processes, Decision trees, Random Forests, Nearest Neighbors).
 - We will be focused on Neural Networks.
- COS 470/570: Introduction to Artificial Intelligence Searching
 - Searching for solutions
 - Neural Networks are more about pattern matching.
- COS 498/598: Explainable AI (Dr Chaofan Chen) ECE 533: Spring
 - Focuses on explainability of Neural Networks

3 Syllabus

Linked here: [Syllabus](#)